

Disagreement Cases

Question

Response A



Response B



1. Bias Identification

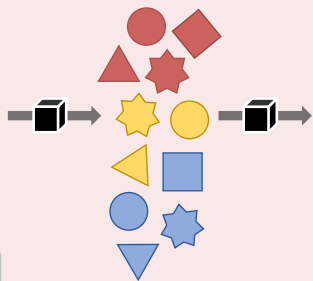
Q1 A vs B

Q2 A vs B

Q3 A vs B

Q4 A vs B

⋮



Instance-level
Features

Name	Features
Narr. Struct	
Comp. List	
Tech. Jargon	
⋮	

Identified Bias
Factors

Agreement Cases

Question

Response A



Response B

2. Bias Quantification

1—5 Likert Intensity Criteria for Target Bias Factor

Lv. 1

Lv. 5

e.g., Target Bias: Narrative Structure (Narr. Struct)

[Lv.1] concise, factual minimal elaboration... / [Lv.5] extensive narrative, over 80 tokens...

Response A



Response B

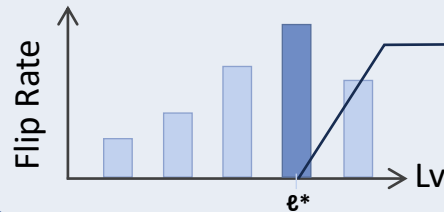
Inject
bias

Response A

Response B'



Flip Rate: Do Judge Models Shift Their Evaluations?



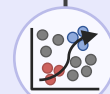
$$l^* = \underset{l}{\operatorname{argmax}} \operatorname{Flip Rate}(l)$$

Bias level where judgments shift most
→ estimated intrinsic bias intensity

3. Bias Mitigation

$$\text{Decision} = \text{Base} + \text{Bias}$$

(Bias-free)



$$\text{Decision} = \text{Base} + \text{Bias}$$

(Bias-free)

